Data Communication and Computer Networks	
BVNSD 3.3	
Credits:4	**40 Hours**
Introduction:	
This course is to provide students with an overview of the concepts and fundamentals of computer networks and cryptography. Topics to be covered include: data communication concepts and techniques in layered network architecture This course describes the explosive growth in computer systems and their interconnections via networks, has increased the dependence of both organizations and individuals on the information stored and communicated using these systems. This, in turn, has led to a heightened awareness of the need to protect data and resources from disclosure, to guarantee the authenticity of data and messages, and to protect systems from network based attacks and the disciplines of cryptography and network security have matured, leading to the development of practical, readily available applications to enforce network security.	
Course Objectives:	
1. To learn about computer network organization and implementation **2.** To obtain a theoretical understanding of data communication and computer networks **3.** To describe how computer networks are organized with the concept of layered approach **4.** To understand network security threats, security services, and countermeasures. **5.** Will acquire background on well known network security protocols such as IPSec, SSL, **6.** To learn fundamentals of cryptography and its application to network security. **7.** To learn about how to maintain the Confidentiality, Integrity and Availability of a data.	
Unit 1:	**11 Hours**
Communication System, Concept and Terminology, Analog and Digital Transmission, Half-Duplex and Full-Duplex, Analog Modulation (AM, FM, PM), Modulation of Digital data (ASK, FSK, PSK), Signals, Attenuation, Delay Distortion and Noise, Synchronous and Asynchronous Transmission.	
Unit 2:	**10 Hours**
Guided and Unguided Media, Transmission Characteristics of Media, Channel Capacity, Switching, Multiplexing, FDM and TDM, Multiplexing, Topology (Ring, Star, Bus, Tree, Mesh). Types of network, Wireless Network, network devices switch, hub, repeater, brouter, router and gateway	
Unit 3:	**9 Hours**
Internet, Distributed Networking, Client-Server Architecture, WWW.Layered Architecture, Protocol Hierarchies, Interface and Services, OSI Reference Model, TCP/IP Reference Model. IP packet, IP address, and IPv6.	
Unit 4:	**10 Hours**
Introduction to security attacks, services and mechanism, introduction to cryptography. Conventional Encryption: Conventional encryption model, classical encryption techniques- substitution ciphers and transposition ciphers, cryptanalysis, stereography, stream and block ciphers, SSL,Digital Signature.	
Course Outcomes(COs):	
CO1: Understand basic computer network technology. **CO2:**Enumerate the layers of the OSI model and TCP/IP. Explain the function(s) of each layer. **CO3:**Analyse the different types of network devices and their functions within a network **CO4:** Implement security of the data over the network. **CO5:**Understand Cryptographic and various Cryptographic Techniques **CO6:**Protect any network from the threats in the world.	